Computer-Based Data Processing

Lab 4 – Linking and Querying Data



Objectives

In the previous labs, we worked mainly with a single table: we transformed data, built pivot tables, and used optimisation tools.

However, data are not always stored in a single table.

In this lab, we will work with several tables describing the **2022 Football World Cup**. The aim is to learn how to:

- understand the role of a **shared identifier** across several tables;
- look up information in another table;
- automatically enrich one table using another;
- filter and sort the resulting data;
- count observations and calculate statistics using one or more criteria.

Download and open the file:

`2022_world_cup.ods`

The workbook contains three sheets:

- `groups`: the teams, their groups, and their FIFA rankings;
- `squads`: the players selected for the tournament;
- `matches`: the World Cup matches.

1 Several tables, a shared identifier

Start by examining the three sheets.

A team such as *France* appears in several tables, but the information associated with it differs.

For example:

squads		groups
France	⟶	France's group
France	⟶	France's FIFA ranking

The `Team` column therefore allows us to link these two tables.

Key idea: information shared by several tables can be used to link them. Here, the team name plays this role.

Your first lookup

Go to the `squads` sheet.

Add two new columns to the right of the table:

`Group` `FIFA Ranking`

We want to fill these columns automatically using information from the `groups` sheet.

The `VLOOKUP` function searches for a value in the first column of a range and retrieves information from the same row.

Its general form is:

```
=VLOOKUP(value; range; column_number; 0)
```

The last argument, `0`, specifies that we want an **exact match**.

In the first data row, build a formula that:

1. takes the name of the player's team;
2. looks up this team in the `groups` sheet;
3. returns the corresponding group.

For example, if the table in the `groups` sheet occupies columns `A:C`, the formula will have this form:

```
=VLOOKUP(team_cell; groups.$A$2:$C$33; 2; 0)
```

Adapt the formula to the cells in your file.

Then fill the formula down for all players.

Retrieving another piece of information

Now use the same approach to fill the `FIFA Ranking` column.

The lookup is the same, but the information to retrieve is in a different column of the `groups` table.

Once you have written the formula, fill it down for all players.

You have just **enriched one table using another** without manually copying the information.

2 Querying the enriched data

The `squads` sheet now contains information about both the players and their teams.

We can therefore query it.

Filtering and sorting

Enable AutoFilter for the table.

Use it to perform the following tasks in turn:

- display only the players from one team;
- display only goalkeepers;
- display players belonging to a particular group;
- sort players from youngest to oldest;
- sort teams by their FIFA ranking.

Then try combining several criteria. For example, display only players in a particular position who belong to a particular group.

Counting using a criterion

The `COUNTIF` function counts cells that meet a condition:

```
=COUNTIF(range; criterion)
```

Use it to calculate:

- the number of players in the France team;
- the number of goalkeepers;
- the number of players belonging to group A.

Next, instead of writing the criterion directly in the formula, place it in a cell.

For example:

```
=COUNTIF(team_range; K2)
```

By changing only `K2`, you can then obtain the number of players in any team.

3 Combining several criteria

We can also ask questions that require several conditions.

The `COUNTIFS` function counts rows that meet several criteria simultaneously:

```
=COUNTIFS(range1; criterion1; range2; criterion2)
```

Use it to count, for example:

- goalkeepers in the France team;
- forwards belonging to group C.

Similarly, `AVERAGEIF` calculates an average using a criterion:

```
=AVERAGEIF(criteria_range; criterion; values_range)
```

Build a small analysis area where you can enter a team name in a cell and automatically display:

- its number of players;
- the average age of its players;
- its FIFA ranking.

Change the selected team and check that the results update automatically.

4 Linking another table

We will now apply the same approach to the `matches` sheet.
This table includes the columns:

`Home Team` `Away Team`

but does not contain the teams' FIFA rankings.
Add two columns:

`Home FIFA Ranking` `Away FIFA Ranking`

Using what you have done so far, write two formulas to look up this information in the `groups` sheet.
This time, no complete formula is provided. Identify for yourself:

- the value to look up;
- the table to search;
- the column containing the information to retrieve.

Fill the formulas down through the entire table.

When lookups fail

Examine the results carefully.
Some rows may produce an error indicating that the lookup value was not found.
In particular, examine the values of `Home Team` and `Away Team` in these rows.
In the knockout stages, some teams are identified by codes such as:

`1A`, `2B`, `W49`, ...

Try looking up one of these values in the `groups` sheet.

Important: a lookup between two tables can only work if the **value used to link them actually exists in both tables**.

5 Building an automatic team profile

We will finish by creating a simple interface for looking up team information.
Create a new sheet called:

`team`

Enter a team name in a cell, for example:

`France`

Then build a profile that automatically displays:

Team	France
Group	...
FIFA ranking	...
Number of players	...
Average age	...

Use the functions covered in this lab so that all the information depends only on the cell containing the team name.

Change this cell to `Brazil`, `Japan`, `Morocco`, etc.

The entire profile should update automatically.

Further practice

If you have finished, choose an additional analysis.

For example, you can:

- identify the team with the highest average age;
- count the players in a particular team who have already scored for their national team;
- compare the number of players in each position between two teams;
- add each team's group to the `matches` sheet.

Review

Throughout these four labs, we have used Calc in different ways:

	Objective	**Main tools**
Lab 1	Transform data	Formulas
Lab 2	Summarise data	Pivot tables
Lab 3	Optimise	Goal Seek, Solver
Lab 4	Link and query	Lookups, criteria, filters

A spreadsheet can therefore be used not only to perform calculations, but also to organise several tables and establish relationships between them using shared information.